

Memory Architecture & Addressing Principles

The load/store model, address computation, data sizing,
and memory hierarchy interactions in AArch64

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Computer Systems Architecture — AArch64 Reference Platform

Topic 5 Outline

1. Memory Access and Computation

- Why AArch64 separates memory transfers from arithmetic operations.
- How values move between memory and the register file.

2. Data Types, Widths, and Conversions

- Handling bytes, halfwords, words, and doublewords cleanly.
- Choosing zero-extension or sign-extension when loading sub-word data.

3. Address Calculation

- Immediate offsets, pre/post-indexing, and scaled array indexing.
- How each addressing form changes, or does not change, the base register.

4. Applied Access Patterns

- Practical array traversal and indexed-array access patterns.
- Choosing the addressing form that matches the data-access pattern.

PART 1

The Load/Store Architectural Boundary

The Fundamental Memory Challenge

- Registers are fast but limited in number; memory holds much more data but is accessed through explicit transfer instructions.
- **Architectural Question:** How should the instruction set organize data movement between memory and the register file?
- AArch64 uses a strict **load/store discipline**: ALU instructions operate on registers, while load/store instructions access memory.

The Architectural Rule

ALU instructions process *only* register operands. Memory access is performed by load/store-family instructions.

Separating Computation from Data Movement

Architectural Separation

- Arithmetic instructions do not directly access data memory.
- Load instructions copy values from memory into registers.
- Store instructions copy register values back to memory.

Typical Data Flow

- 1 **Load:** bring data into registers.
- 2 **Compute:** operate on register values.
- 3 **Store:** write the result back if needed.



Addressing Memory: Bytes, Words, and Doublewords

- Memory is a linear sequence of **8-bit bytes**; each byte has its own address.
- The instruction/access size determines the transfer width:

Data Type	Width	Load Syntax	Store Syntax
Byte	8-bit	<code>ldrb wt, [xn]</code>	<code>strb wt, [xn]</code>
Halfword	16-bit	<code>ldrh wt, [xn]</code>	<code>strh wt, [xn]</code>
Word	32-bit	<code>ldr wt, [xn]</code>	<code>str wt, [xn]</code>
Doubleword	64-bit	<code>ldr xt, [xn]</code>	<code>str xt, [xn]</code>

The base register (xn) holds the memory pointer enclosed in brackets: [xn].

PART 2

Data Sizing & Extension Rules

Sub-Word Loads and Extension

- Registers are 64 bits wide, but memory values may be only 8, 16, or 32 bits.
- When a smaller value is loaded, the processor must fill the remaining upper bits.
- The correct choice depends on whether the memory value should be treated as **unsigned** or **signed**.

Zero-Extension

Use for unsigned data. Upper bits are filled with 0s.

Sign-Extension

Use for signed data. The sign bit is copied into the upper bits.

Sign vs. Zero Extension: Side-by-Side

Memory Byte at Address x1 = 0x80 (Binary: 1000 0000)

The same 8-bit pattern means either unsigned 128 or signed -128, depending on the load instruction.

Instruction	Value in x0	Meaning
ldrb w0, [x1]	0x00000000_00000080	Zero-extends the byte; interpreted as unsigned +128.
ldrsb x0, [x1]	0xffffffff_ffffff80	Sign-extends the byte; preserves signed value -128.

Using a zero-extending load for signed negative data can turn a negative value into a large positive one.

Storing Data: Truncation Rules

- Stores move data from a register into memory.
- The instruction width determines how many bytes from the register are written:

Instruction	Bits Read from Register	Bytes Written to Memory
<code>strb wt, [xn]</code>	Bits [7:0]	Writes exactly 1 byte
<code>strh wt, [xn]</code>	Bits [15:0]	Writes exactly 2 bytes
<code>str wt, [xn]</code>	Bits [31:0]	Writes exactly 4 bytes
<code>str xt, [xn]</code>	Bits [63:0]	Writes full 8 bytes (doubleword)

Storing a byte or halfword modifies *only* the targeted memory locations; neighboring bytes are left unchanged.

PART 3

AArch64 Addressing Modes

Addressing Modes

- An **addressing mode** is how a load/store instruction forms the memory address it will access.
- The resulting address is the **effective address**: the actual byte address used by the load or store.

Access Pattern	Addressing Form	C Pointer Pattern	Typical Use
Fixed byte offset from a base pointer	<code>[xn, #imm]</code>	byte offset from <code>ptr</code>	Struct fields, known offsets
Step through 64-bit elements	<code>[xn], #8</code>	<code>*ptr++</code> for <code>uint64_t *ptr</code>	Array traversal
Base plus scaled index	<code>[xn, xm, lsl #3]</code>	<code>arr[i]</code> for 64-bit elements	Indexed array access

AArch64 offsets are byte counts. C pointer arithmetic scales by the size of the pointed-to type.

Mode 1: Base Register with Immediate Offset

- Adds a fixed byte offset to a base address without modifying the base register.
- **Syntax:** `ldr xt, [xn, #offset]` $\Rightarrow$ $EA = x_n + \text{offset}$.



Example: Reading Values at Fixed Offsets

Assembly	C-like Pattern	Meaning
<code>ldr x1, [x0, #0]</code>	<code>*ptr</code>	64-bit value starting at x0.
<code>ldr x2, [x0, #8]</code>	<code>*(ptr + 1)</code>	For a 64-bit pointer, reads the next element, 8 bytes from x0.
<code>ldr w3, [x0, #16]</code>	<code>p[4]</code>	For <code>int32_t *p</code> , reads the 32-bit value starting 16 bytes from x0: four 32-bit elements from the base.

Mode 2: Pre-Indexed Addressing (Update Before Access)

- Updates the base register **before** reading or writing memory.
- **Syntax:** `ldr xt, [xn, #offset]!` (Exclamation mark indicates update).

Step-by-Step Action

- 1 $x_n \leftarrow x_n + \text{offset}$ (Update pointer)
- 2 Access memory at the **new** address in x_n .

Primary Use Case: Access After Pointer Adjustment

`ldr x0, [x1, #-8]!` $\approx$ `*--ptr for uint64_t *ptr`

Subtracts 8 bytes first, then loads from the updated address.

Mode 3: Post-Indexed Addressing (Update After Access)

- Reads or writes memory at the current address, then updates the base pointer **afterwards**.
- **Syntax:** `ldr xt, [xn], #offset` (Offset written outside the brackets).

Step-by-Step Action

- 1 Access memory at the **current** address in x_n .
- 2 $x_n \leftarrow x_n + \text{offset}$ (Advance pointer)

Primary Use Case: Array Stepping

`ldr x0, [x1], #8` $\approx$ `*ptr++` for `uint64_t *ptr`

Loads from the current pointer, then advances by one 8-byte element.

Comparing the Three Offset Modes

Assumption: the C pointer refers to 64-bit elements, so one C element step equals 8 bytes.

Mode	Syntax	C Pattern	Address Accessed	Base After
Standard Offset	[xn, #8]	*(ptr + 1)	$x_n + 8$	Unchanged
Pre-Indexed	[xn, #8]!	+++ptr	$x_n + 8$	Updated
Post-Indexed	[xn], #8	*ptr++	x_n	Updated

Assume x1 = 0x1000 Initially

ldr x0, [x1, #8]	Reads from 0x1008;	x_1 remains 0x1000
ldr x0, [x1, #8]!	Reads from 0x1008;	x_1 becomes 0x1008
ldr x0, [x1], #8	Reads from 0x1000;	x_1 becomes 0x1008

Mode 4: Register Offset with a Scaled Index

Purpose

Use a base address plus an index register to access `arr[i]`; the index is explicitly scaled by the element size.

```
ldr x0, [x1, x2, lsl #3]
```

- `x1`: base address; `x2`: element index.
- `lsl #3`: multiply the index by 8 for 64-bit elements.
- Effective address: $x_1 + (x_2 \times 8)$.

Element Type	Element Size	Index Scale	Example Load
Byte	1 byte	none	<code>ldrb w0, [x1, x2]</code>
Halfword	2 bytes	<code>lsl #1</code>	<code>ldrh w0, [x1, x2, lsl #1]</code>
Word	4 bytes	<code>lsl #2</code>	<code>ldr w0, [x1, x2, lsl #2]</code>
Doubleword	8 bytes	<code>lsl #3</code>	<code>ldr x0, [x1, x2, lsl #3]</code>

Here, scaling is programmer-visible: scale the index by the element size, then add it to the base address.

PART 4

Multi-Register Transfers & Applied Access Patterns

Paired Transfers: `ldp` and `stp`

- `ldp` and `stp` transfer **two adjacent registers** with one architectural instruction.
- They are useful when nearby values are loaded or stored together.

Instruction	Action	Typical Use
<code>ldp x0, x1, [x2]</code>	Load two 64-bit values	Adjacent data values
<code>stp x0, x1, [x2]</code>	Store two 64-bit values	Adjacent output values
<code>ldp x2, x3, [x1, #16]</code>	Load pair at an offset	Copying or scanning data

Paired transfers reduce instruction count when the program naturally works with adjacent values.

Worked Pattern 1: Stepping Through an Array

Goal: Sum n 64-bit integers, with `uint64_t *ptr`

C pattern: `while (n--) sum += *ptr++;`

AArch64 Implementation (Using Post-Indexed Addressing)

```
// x0 = array pointer, x1 = count n
sum_loop:
    mov x2, xzr          // sum = 0
    cbz x1, .Ldone       // exit if count == 0

.Lbody:
    ldr x3, [x0], #8     // load element and advance by 8 bytes
    add x2, x2, x3       // sum += element
    sub x1, x1, #1       // count-
    cbnz x1, .Lbody      // loop if count != 0

.Ldone:
    mov x0, x2           // return sum in x0
    ret
```

Post-indexing combines the load and pointer advance in one instruction.

Worked Pattern 2: Indexed Array Search

Goal: Check Element at Index i in Array

C pattern: `arr[i]`

AArch64 Implementation (Using a Shifted Register Offset)

// x0 = array pointer, x1 = index i, x2 = target value

`find_element:`

```
    ldr x3, [x0, x1, lsl #3]    // load arr[i] using base + (i * 8)
    cmp x3, x2                  // compare arr[i] with target
    cset w0, eq                  // w0 = 1 if match, else 0
    ret
```

Uses a shifted register offset to form the address of `arr[i]` and load it in one instruction, without modifying the base pointer `x0`.

Addressing Mode Selection Guide

Programming Scenario	C Pattern	Addressing Mode	Example Instruction
Accessing a known byte offset	byte offset from base	Base + immediate offset	<code>ldr x1, [x0, #8]</code>
Loading next 64-bit element	<code>*ptr++</code> for <code>uint64_t *ptr</code>	Post-indexed update	<code>ldr x2, [x0], #8</code>
Stepping backward 64-bit data	<code>*--ptr</code> for <code>uint64_t *ptr</code>	Pre-indexed update	<code>ldr x2, [x0, #-8]!</code>
Accessing a 64-bit indexed element	<code>arr[i]</code>	Scaled register offset	<code>ldr x2, [x0, x1, lsl #3]</code>
Moving two adjacent values	<code>ptr[0], ptr[1]</code>	Paired transfer	<code>ldp x2, x3, [x0]</code>

Common Memory Access Mistakes

- **Wrong Sizing:** Using `ldr xt` instead of `ldr wt` on an array of 32-bit integers reads too many bytes.
- **Missing Sign Extension:** Using `ldrb` instead of `ldrsh` on signed byte data changes negative values into positive ones.
- **Incorrect Shift Scale:** Using `lsl #2` instead of `lsl #3` on an array of 64-bit values computes the wrong address.
- **Accidental Base Update:** Using pre-indexed or post-indexed addressing when the original pointer still needs to be preserved.

SUMMARY & ASSESSMENT

Review and Self-Test Questions

Topic 5 Summary

- **Load/Store Discipline:** ALU instructions operate on registers; load/store-family instructions move data between registers and memory.
- **Data Widths & Extensions:** The access width determines the bytes transferred. Loads use zero-extension or sign-extension to fill upper bits.
- **Stores & Paired Transfers:** Narrow stores write only the low bits. `ldp/stp` transfer two values at consecutive memory locations.
- **Core Addressing Modes:**
 - Standard offset (`[xn, #imm]`): byte offset, base unchanged.
 - Pre-indexed (`[xn, #imm]!`): update base, then access memory.
 - Post-indexed (`[xn], #imm`): access memory, then update base.
 - Shifted register offset (`[xn, xm, lsl #3]`): shift index to form a byte offset, then add to base. Base unchanged.
- **C vs. Assembly:** C pointer arithmetic is element-scaled; AArch64 immediate offsets are byte displacements.

Self-Test Questions: Data Transfers & Sizing

- ❶ **Load/Store Discipline:** A 64-bit value is stored at the address in x1. Write a load, an add, and a store to increase it by 1, using x0 as a temporary register.
- ❷ **Zero-Extension vs. Sign-Extension:** The byte at x1 is 0xfe. For each load below, give the full 64-bit pattern in x0 and its decimal value. Which load preserves the signed 8-bit value -2?

`ldrb w0, [x1]` `ldrsw x0, [x1]`

- ❸ **Store Width:** If w0 contains 0x12345678, which bits does `strb w0, [x1]` store, how many bytes does it write, and does it change neighboring bytes?
- ❹ **Paired Transfers:** If $x2 = 0x4000$, which addresses supply the two 64-bit values loaded by `ldp x0, x1, [x2]`? How many bytes are read, and does x2 change?

Self-Test Questions: Addressing & Array Traversal

- ➊ **Offset Modes:** Treat each instruction independently, starting with $x1 = 0x3000$. Give the address read and the final value of $x1$ for each:

```
ldr x0, [x1, #8]    ldr x0, [x1, #8]!  
ldr x0, [x1], #8
```
- ➋ **Shifted Register Offset:** A `uint32_t` array starts at $x0$, with index i in $x1$. Write one instruction to load `arr[i]` into $w2$. Explain the shift and whether either address register changes.
- ➌ **Summation Loop:** In Worked Pattern 1, which instruction implements `*ptr++`? Why does $x0$ advance by 8 while $x1$ decreases by 1? What does the initial `cbz` do when $n = 0$?
- ➍ **Backward Traversal:** For `uint64_t *ptr` in $x1$, write one instruction to load `*--ptr` into $x0$. Which addressing mode does it use, and when does the pointer change?

Looking Ahead

Next Lecture Preview

Our Next Topic: AArch64 Subroutines, Calling Conventions, & the Stack

- **Subroutine Calls:** Using `bl`, `blr`, and `ret`.
- **Calling Convention:** Passing arguments, returning values, and preserving registers.
- **Stack Frames:** Applying stack alignment, saved registers, and frame layout in real subroutines.